

Inventory Optimization

A Strategic Framework for Zebra Technologies' Global Spare Parts Network

MS Business Analytics Capstone | Final Report

McCombs School of Business | The University of Texas at Austin

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Executive Summary

Zebra Technologies operates a global service network of more than 50 repair locations responsible for keeping the company's enterprise hardware (barcode scanners, RFID readers, mobile computers, industrial printers) operational for over 80% of Fortune 500 customers. Maintaining customer turnaround commitments depends on accurately stocking spare parts at each depot, but Zebra's current planning process relies on a static heuristic that is structurally unable to balance service performance against capital efficiency.

This capstone project delivers a strategic framework for inventory optimization across Zebra's global service network. We analyzed stock levels, demand patterns, and supplier performance for 21 core service locations using more than 200,000 internal replenishment records and 13 months of consumption history. Our objective was to determine the optimal inventory level for each spare part / location combination, evaluate whether Zebra can maintain or improve customer satisfaction at lower network-wide inventory, and surface the operational dependencies that must be addressed before operating at those leaner levels.

Our central finding is that Zebra's static ABC segmentation is driving substantial capital bloat: 'C'-rated and 'Unassigned' parts together consume roughly 85% of total inventory space despite their low criticality. We replaced the heuristic with a Hybrid Inventory Optimization Pipeline that routes each part to the most appropriate statistical model: a Poisson Generalized Linear Model for parts with a traceable installed-base footprint, and a time-series fallback with a 95% service-level Z-score buffer for the remainder. The pipeline is unified by an automated decision gate that selects the right model per part and writes mathematically optimal re-order points and safety stocks to a Power BI dashboard for planner consumption.

Validation against Zebra's current model on a representative depot showed measurably lower forecast error (MAE), correction of a positive predictive bias toward over-stocking, and an inventory turnover rate roughly three times higher than the baseline. Network-wide, our recommended targets reduce required inventory month-over-month while preserving the 95% material fill rate. Most locations were over-target relative to our model, and we observed no positive correlation between higher inventory levels and on-time shipment performance: the over-stocked depots were not the best performers on customer satisfaction, undermining the implicit assumption behind the current heuristic.

Two operational dependencies must be addressed before Zebra can safely operate at these leaner levels. First, internal DC-to-Depot transfers exhibit a late-delivery rate with a heavily right-skewed lead-time distribution extending past 80 days; this internal variance, not external vendor performance, is the binding constraint on depot stockouts. Second, master data gaps in the Bill of Materials limit Poisson coverage on certain depots, forcing the remainder into a more conservative time-series fallback. We recommend Zebra adopt a phased rollout that combines our Poisson-driven targets at smaller depots with a tuned version of the existing approach at major regional hubs, while in parallel investing in internal SLA enforcement and BOM master data cleanup.

1. Business Context and Problem Statement

1.1 Company Overview

Zebra Technologies is a leading American company in data capture and automation solutions, specializing in barcode scanning, RFID, mobile computing, and machine vision. Its products enable real-time visibility across business operations and are widely used in retail, healthcare, manufacturing, and logistics. More than 80% of Fortune 500 companies, including Dell Technologies, AT&T, and Microsoft, rely on Zebra to digitize and modernize frontline workflows.

The scope of this project focuses on Zebra's hardware service network. Zebra operates more than 50 service locations where customer devices are sent to be repaired and triaged. To meet contractual turnaround-time and quality obligations, service planners must accurately predict demand and supply so that the correct number of spare parts is maintained at each location. The major costs in this network are materials, shipping, and repairs, and the network's inventory directly determines Zebra's ability to honor repair commitments while controlling the working capital tied up in spare parts.

1.2 Problem Statement

Zebra's current spare parts planning process uses a static ABC heuristic that assigns arbitrary, fixed weeks-of-supply targets: for example, X number of weeks for A-rated parts, Y number weeks for B-rated parts, and Z number of weeks for C-rated parts. This approach has two structural weaknesses: it ignores the empirical demand variance of each individual part, and it applies the same blanket logic across geographically and operationally distinct depots whose lead-time risk profiles differ substantially. The result is a network that simultaneously over-stocks low-criticality items and under-buffers parts whose volatility actually drives customer satisfaction.

This project provides a strategic framework for inventory optimization that analyzes stock levels and performance across Zebra's global service network. The work was structured around three business questions provided by the sponsor:

1. What is the optimal inventory level for each spare part/location combination?
2. Can Zebra improve or maintain current customer satisfaction scores and material fill rates with a lower network-wide inventory level?
3. What are the key dependencies that must be addressed for Zebra to operate under optimal inventory levels?

1.3 Business Value and Expected Outcomes

The framework developed in this project delivers value across four dimensions:

- **Cost reduction and avoidance:** lower total network-wide inventory value, reduced holding and obsolescence costs, fewer emergency shipments triggered by stockouts, and avoided purchases of slow-moving parts.
- **Efficiency gains:** streamlined replenishment planning, reduced reactive inter-location transfers, and faster cycle times because planners no longer firefight imbalances created by static targets.
- **Risk reduction:** earlier identification of depot-level stockout risk, explicit quantification of supplier and internal lead-time variability, and contingency visibility for tariff or supply disruptions.
- **Improved decision quality and customer experience:** data-driven, defensible inventory targets per part/location, support for cost-vs-service trade-off analysis, and protection of fill rate and on-time service performance, which are the metrics that ultimately drive customer retention.

2. Data Sources and Preparation

2.1 Data Environment

All analysis was conducted within Zebra's secure IT environment. Our team's primary access point was a suite of curated Power BI reports and semantic models that source, join, and aggregate data from underlying tables, providing an abstraction layer that ensured the analysis was built on validated, business-approved sources.

2.2 Primary Data Sources

Four Power BI reports anchored the analytical model. Each draws from specific granular BigQuery tables and addresses a different facet of the inventory optimization problem:

- **Consumption Planning Tool Report:** the foundation for demand forecasting. It supplies monthly consumption quantities for each part at each service depot, drawn primarily from one table. Two other tables were joined on the primary table to retrieve more information on the spare parts.
- **Supplier Commit Metrics (ASN) Report:** the source for end-to-end purchase order history and lead-time analysis. Three tables were looked at for our analysis.
- **Inventory On-Hand Report:** snapshot of current on-hand inventory levels and product information. Two tables were used from this report and were the main basis of our analysis.
- **Service Open Sales Orders Report:** visibility into customer orders by flow status. This data is essential for measuring material fill rate and backorder levels.

To answer Business Question 2 we needed additional data sources to evaluate customer satisfaction and actual Zebra Inventory levels. To do this we used 4 additional Big Query tables and 1 Power BI report:

- **Additional Tables Not Found in Recommended Reports:** table containing various repair related information. This was used primarily in evaluating customer satisfaction metrics, and the Power BI report was used to validate the table by leadership. Another table containing information on shipments. The final table containing customer information was used with the previous table to evaluate what customers' various shipments were being sent to.

2.3 Data Quality Assessment

Two material data quality issues surfaced during initial assessment. The first was mismatched variable documentation. We addressed this by cross-checking each table against actual joins and building our own internal reference sheets to validate which variables were live.

The second issue was unassigned inventory ratings. Zebra's ABCD rating drives current planning logic, but a portion of service parts carried 'Unassigned' labels. Because a meaningful share of the catalog is ungraded, this gap propagates into any model the current conditions rating, and it was a primary reason we replaced the heuristic entirely rather than tuning it.

The third issue was empty and redundant data fields. Several tables used had useful fields that were mostly empty. Many important fields holding important information like customer ship dates or auxiliary information used to calculate various quantity and datetime metrics were empty. We also had large redundancies in several tables with many fields holding the same data, pulled from different databases used by different departments or databases that were discontinued.

2.4 Data Gaps and Workarounds

Two gaps directly affected our ability to answer the business questions. First, no field in the data tables explicitly identified which parts and locations are 'service'. Zebra provided specific filtering requirements for variables in order to bypass this.

Second, customer satisfaction surveys are stored in a separate system to which we did not have access. To proxy for customer satisfaction, we engineered three quantitative variables from 3 different tables. These metrics let us measure on-time shipping behavior and supplier performance, and they provide the optimization targets we use throughout the model.

For addressing missing data fields, we focused on creating tables that analyzed the null values per field to identify the useable data fields we had. We then used feature engineering or identified additional tables and resources to fill in gaps as needed.

2.5 Data Engineering and Preparation

The objective of the data engineering phase was to transform disparate raw tables into a single, unified, feature-rich dataset suitable for inventory optimization modeling. We followed an 'Aggregate-Then-Join' strategy to ensure both computational efficiency and analytical integrity, executed in three connected steps.

Smart-Clean Logic

A programmatic cleaning function (`super_smart_clean_family_name`) corrects systematic ERP errors before any modeling. All temporal data is coerced to a YYYY/MM format using strict date parsing,

guaranteeing join integrity across the 13 trailing months of consumption history. We bypassed an inherent pandas parsing flaw whereby the North American region code 'NA' was being evaluated as a null value, by applying explicit string imputation. Disjointed product naming conventions were unified by stripping and upper-casing string inputs to establish reliable primary keys between attribute_1 and attribute_2.

Aggregation, Joining, and Imputation

We aggregated the demand-history and receipt-history tables by Part Number and Warehouse Location before joining them, deliberately starting from the master list of all products and applying a LEFT JOIN so that every part is preserved in the final analysis even when no demand or lead-time history exists. Missing values introduced by the join were imputed with zero, a business-driven choice rather than a statistical convenience, because in this context a null demand or null lead time genuinely represents 'never sold' or 'never ordered' and should propagate as a zero-safety stock for downstream optimization.

Feature Engineering and Filtering

Several features serve as direct inputs to the optimization model:

- **Lead Time (Days):** calculated as the date difference between Receipt Date and Ship Date on each purchase order line.
- **Average Demand and Demand Volatility:** the central tendency and historical unpredictability of monthly consumption per part / location.
- **Average Lead Time and Lead Time Volatility:** supplier mean performance and consistency, the two inputs needed to size variance-aware safety stock.

Records with non-positive calculated lead times were excluded as data-entry or system-timing errors; including them would have artificially compressed lead-time variance and produced inappropriately low recommended inventory levels.

When evaluating Zebra inventory levels, we did additional data engineering. We ordered by dates and aggregated product number quantities per location and month across the evaluated time relevant to our model. Feature engineering wise we extracted location codes from various location descriptions to join with our other tables. We also computed additional features for customer satisfaction such as overall repair lead times, days late per repair, and percentage of late shipments per location for repair specific information.

3. Exploratory Data Analysis

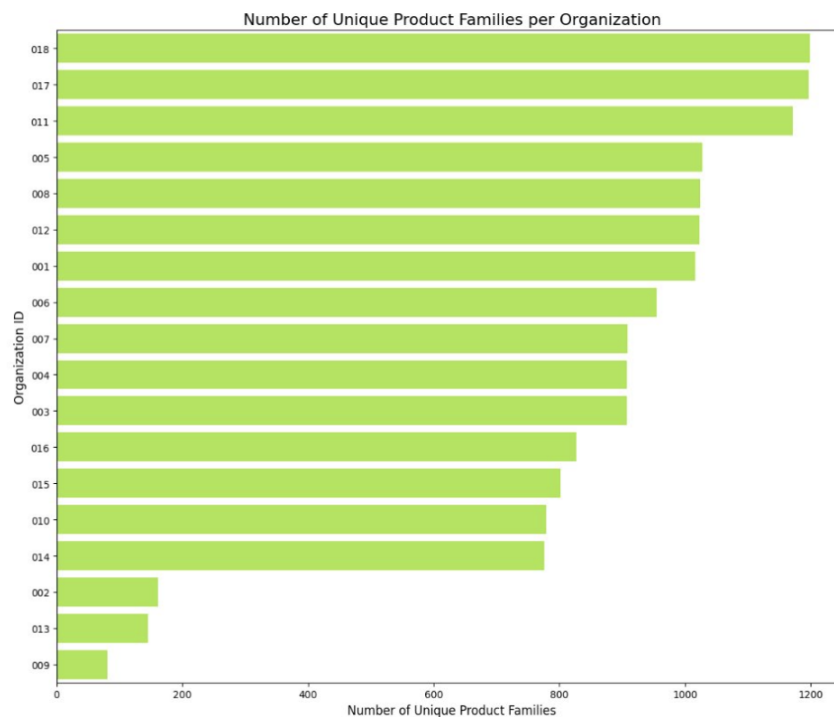
3.1 Source Data Selection

Initial analysis drew on four operational reports (On-Hand, Consumption Planning, Supplier Commit Metrics, and Inbound ASN) to triangulate the data we would need for modeling. After cross-referencing

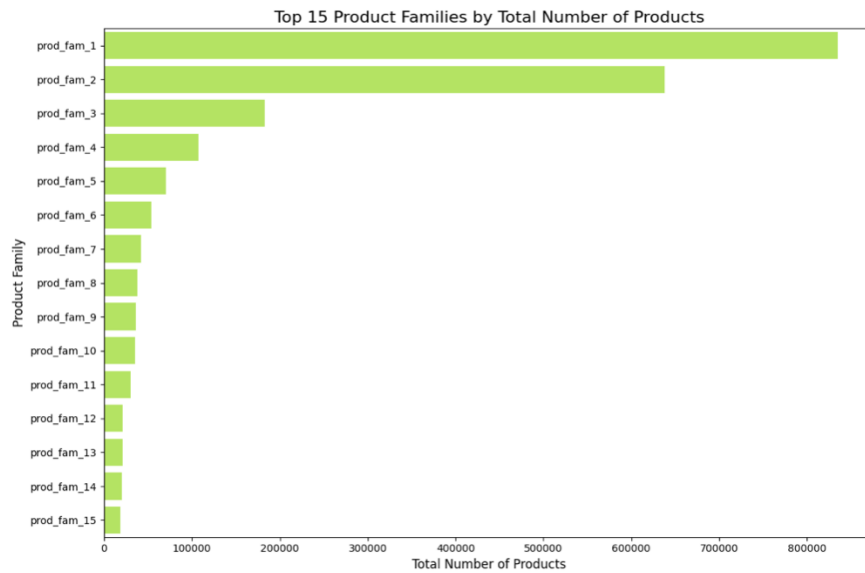
variables and joining behavior, we narrowed the working dataset down to four core tables that supplied demand, supply, on-hand, and order-fulfillment signals at the part/location grain.

3.2 Organization and Product Family Distribution

Across the in-scope organizations, each Organization ID contained a large number of distinct product families, with most carrying around 1,000 unique families, with three organizations carrying close to 1,200 and three at the long tail carrying fewer than 200. The breadth of product families per location was a first signal that depot-level stocking decisions could not be reduced to a small number of representative items, and motivated us to look more deeply into product families as the unit of analysis.



When we then ranked product families by total number of products in inventory, the distribution was sharply concentrated. The top two families accounted for the bulk of product volume (on the order of 800,000 and 600,000 products respectively), while the next thirteen families combined contributed only a fraction of that total. This Pareto-like concentration suggested that a segmented grading system organized around the dominant product families per organization code could produce more meaningful targets than a single global model.



3.3 ABC Segmentation Failure

The most consequential EDA finding concerned Zebra's existing ABC segmentation. Our 13-month consumption analysis showed that 'C'-rated products consumed the majority total inventory space with 'Unassigned' products the second most. Because these are by definition the lowest-criticality parts (or parts that have never been formally graded), the heuristic is allocating the majority of the network's capital to the lowest-impact items. This finding alone justified abandoning the heuristic in favor of a model whose buffers scale with empirical variance.

3.4 Demand Extraction and Manual Cleaning

We extracted monthly demand for each spare part within each organization code, generating long-format tables with columns for product number, product family, organization code, attribute, date, and demand. Joining issues between source tables required iterative manual matching and cleaning before the data was usable for modeling, particularly around product-family identifiers that did not link cleanly across the demand and master tables. The Smart Clean function described in Section 2.5 is a direct outgrowth of the rules we developed during this manual reconciliation.

4. Approach and Methodology

4.1 Why a Hybrid Approach

Spare parts demand at Zebra is highly sparse and intermittent: most parts have many zero-demand months interspersed with rare 'event' months when a unit fails. We initially explored deploying global Machine Learning models such as Random Forest or XGBoost across all parts, but discarded that direction

because standard ML models overfit to zeros and fail to capture the rare-event structure that defines spare parts. Standard regression models such as OLS likewise assume continuous, normally distributed demand that this data emphatically does not exhibit.

The problem structure instead demanded a bifurcated approach. For parts whose failure rate can be tied mathematically to the population of devices in the field, we use a Poisson Generalized Linear Model, the canonical distribution for rare-event counts. For parts whose installed-base linkage is missing or unreliable, we route to a time-series fallback that produces a statistically defensible buffer using only the part's own consumption history. An automated decision gate selects the route per part, allowing the pipeline to handle the entire catalog without manual intervention.

4.2 Pipeline Architecture

The end-to-end pipeline operates in three connected stages.

Stage 1: Data Engineering and Smart Clean

Disparate streams from 20+ tables are standardized through the Smart Clean logic described in Section 2.5: timestamp coercion to YYYY/MM, systemic correction of the 'NA' region-code parsing flaw, and relational mapping of orphaned product numbers to their parent product families. This stage produces the validated primary keys on which all downstream modeling depends.

Stage 2: Automated Demand Routing

An algorithmic decision gate examines each part:

- If a traceable installed base is found ($\text{total_installed_base} > 0$), the part is routed to the Poisson model.
- If no family match exists or BOM linkage is missing, the part is routed to the Time-Series Fallback.
- Both routes are sized to a 95% service-level target, ensuring consistent SLA coverage across the network.

Stage 3: Lead-Time Variance and Final Synthesis

The demand-side output is then merged with empirical lead-time variance from the internal DC-to-Depot replenishment rows. The final deliverables (`_COMPLETE.csv` files per depot) combine demand variance (Poisson or time-series) with supply variance to generate mathematically optimized re-order points and safety stocks per part/location.

4.3 The Poisson Generalized Linear Model

For product families with a non-zero installed-base footprint, we fit a Generalized Linear Model with a Poisson family link using the statsmodels library. The model regresses monthly failure count on total

installed base, capturing the structural relationship that more devices in the field generate proportionally more rare-event repair demand.

The Poisson distribution is explicitly designed to predict the probability of a given number of rare events occurring in a fixed interval, which matches the operational reality of spare part failures far better than continuous distributions. To enforce Zebra's 95% service-level target, the model takes its predicted demand and extracts the 95th percentile using `scipy.stats.poisson.ppf(0.95)`. This guarantees that the depot holds enough inventory to satisfy demand in 95 out of 100 theoretical months. The resulting family-level optimal inventory level is then allocated proportionally to specific part numbers based on each part's historical share of family-level demand.

4.4 The Time-Series Fallback

For parts lacking BOM linkage to the installed base, the pipeline routes to a time-series model based on the part's own consumption history. Safety stock is set at the historical mean monthly demand plus 1.645 standard deviations, where 1.645 is the Z-score corresponding to a 95% service level under a normal approximation. The specific inventory target per part is computed as:

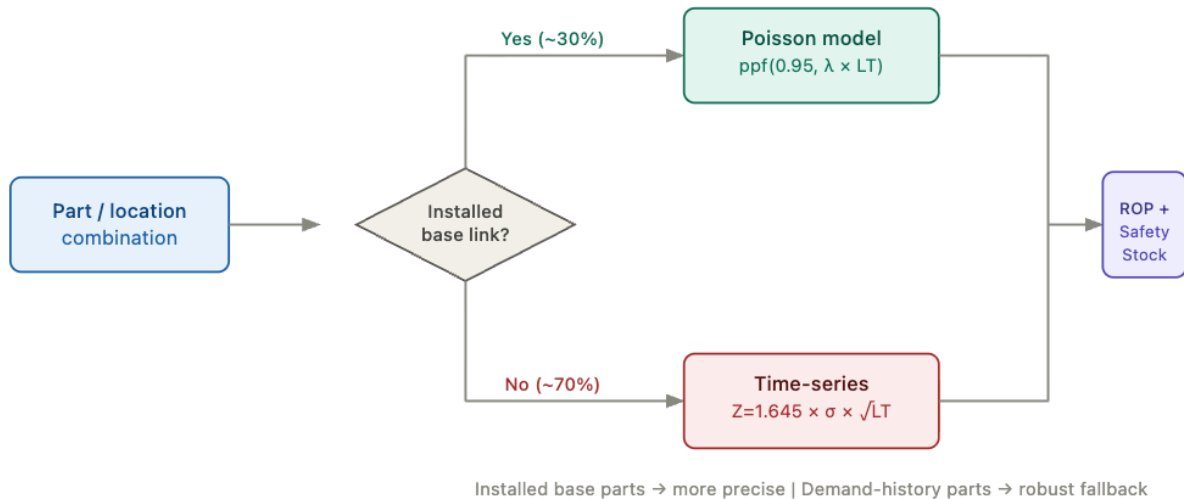
$$\text{Specific Inventory Target} = \lceil \text{Mean} + (1.645 \times \text{Std_Dev}) \rceil$$

This formulation matches the same 95% SLA target enforced by the Poisson route, ensuring consistent risk coverage despite the simpler input data. The fallback is mathematically conservative and directionally reliable, but it cannot anticipate spikes from new product introductions because it depends entirely on trailing history, a limitation we treat explicitly in Section 6.

4.5 Lead-Time Variance Modeling

We abandoned the standard assumption of deterministic supply. Across the 233,588 internal-transfer rows, we computed the empirical delta between Request Date and Depot Arrival Date and characterized the resulting distribution. The mean lead time is 24.5 days, but the distribution is heavily right-skewed: a long tail of late deliveries extending past 80 days for some suppliers, and outliers reaching 150+ days.

Because the distribution is not normal, our safety-stock algorithm uses the empirical Coefficient of Variation rather than assuming a bell curve. This was a deliberate methodological choice that allows the model to accurately capture the extreme right-tail variance observed in practice.



4.6 Output and Dashboard

The final per-part outputs were surfaced through a Power BI dashboard that displays predicted target inventory levels for each part number at each depot. Where unit cost data was available and confirmed, parts are also reclassified into A, B, or C tiers using an 80/15/5 distribution rule on cumulative cost contribution, providing planners with both the optimal level and a value-tier indicator in a single view. A representative row of the output schema looks like:

Product Number	Target Level	Model Used	Location / ABC Rating
Part XYZ	X	Poisson	Location Z / A

4.7 Assumptions and Reasonableness Checks

The methodology rests on three assumptions, each of which we tested against the data:

- **Demand stationarity (partially holds):** standard inventory models assume stable, continuous demand. For spare parts this is violated. We justify the Poisson distribution because it accommodates the non-stationary, intermittent nature of rare part failures better than standard regression.
- **Normal lead-time distribution (violated):** we initially assumed internal DC-to-Depot lead times would be approximately normal. The empirical distribution proved heavily right-skewed with a

long tail extending past 150 days. We responded by sizing safety stock from the empirical Coefficient of Variation rather than imposing a normal assumption.

- **Data integrity and product-family independence (violated):** we initially assumed product numbers would map cleanly to product families. ERP naming inconsistencies broke this assumption. The Smart Clean mapping layer resolves it programmatically, so that final outputs are built on validated, independent family lines.

5. Validation and Results

5.1 Validation Approach

We used out-of-time validation rather than a random train/test split, because random splits leak future information in time-series data and are inappropriate for inventory simulation. Models were trained on earlier operational months (early 2025) using over 1,000 time-series parts and 60 Poisson families in iterative batches, then evaluated against actual chronologically ordered consumption logs from late 2025 through Q2 2026. The ultimate test of an inventory model is not point-in-time accuracy but whether the recommended buffer would have prevented a historical stockout, and an out-of-time simulation is the cleanest way to measure that.

5.2 Metrics and Baseline

We deliberately avoided standard accuracy metrics such as RMSE that are misleading for sparse data, where predicting zero is usually right but operationally useless on the months when a part actually fails. Our primary metric was the 95% material fill rate: the model is judged on its theoretical ability to absorb the empirically observed late-delivery shock without stocking out the depot. The baseline is Zebra's current static ABC heuristic that assigns X weeks of supply to A-rated parts, Y weeks to B-rated parts, and Z weeks to C-rated parts.

5.3 Comparative Performance: Business Question 1

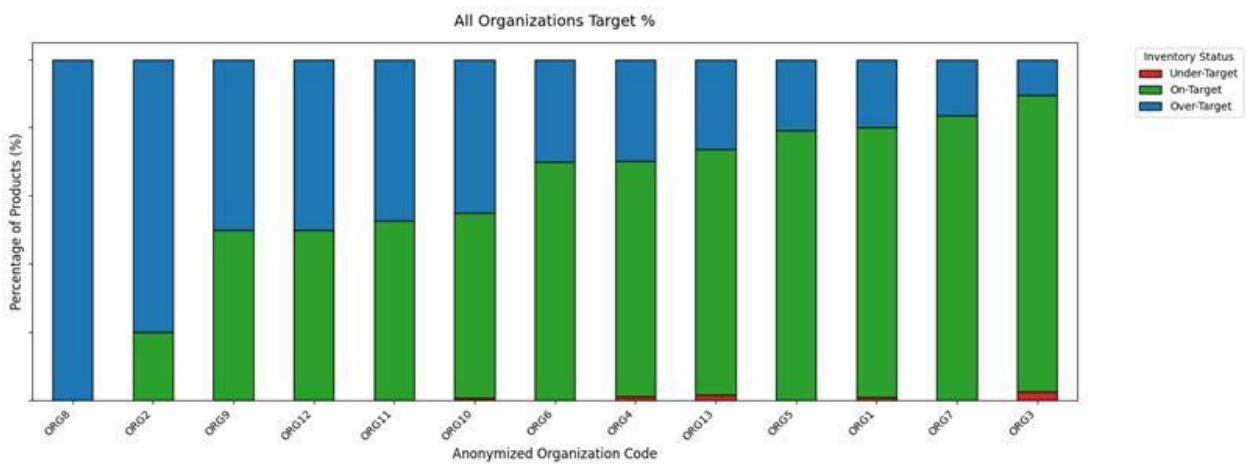
On a representative individual depot in April, the new model outperformed the current model on every measured dimension. The comparison is summarized below:

Metric	Zebra's Current Model	Our New Model
Predictive Accuracy (MAE)	Lower	Higher
Predictive Bias	Positive (over-stocks)	Negative (lean)
Inventory Turnover Rate	1×	3×

Three concrete implications follow. First, the new model has measurably lower forecast error, meaning planners can place capital with greater confidence. Second, the bias correction matters: Zebra's current model is positively biased, meaning it systematically over-predicts demand and over-stocks; our model corrects that bias and brings inventory closer to true expected demand. Third, the threefold improvement in inventory turnover translates directly to reduced excess-inventory costs and reduced overall financial risk on the balance sheet.

5.4 Network-Wide Comparison: Business Question 2

We compared our model's predicted inventory levels per service product/location to Zebra's actual inventory levels for April. Most locations were on-target or substantially over-target relative to our recommendations; only a small fraction of parts at a few locations were under-target. In parallel, we evaluated customer satisfaction by comparing actual ship dates to promised ship dates and assessing average lead times by location.



The pairing of these two views produced one of the most operationally important findings of the project: most locations were close to on-time (less than 5 days late on average), while two locations shipped 20+ days late. Critically, the highly over-stocked locations performed best on customer-satisfaction proxies, but the late locations were mostly already on-target or partially over-stocked. Higher overstocking levels when compared to our model were correlated with better on-time performance. This shows the room for improvement our model would provide. Certain highly overstocked locations could reduce excess part levels and maintain customer satisfaction. The other, poorly performing locations that scored mostly on-target with respect to our model would not be able improve their customer metrics with reducing stock levels,. Additionally, due to their current on-target score with our more conservative model and the magnitude of lateness, they are likely suffering from separate procedural issues resulting in these repair delays.



5.5 Validation Against Baseline

The out-of-time simulation showed that the static ABC heuristic is structurally inefficient. Under the current system, 'C'-rated and 'Unassigned' parts consume majority of total inventory space despite their low criticality. The Hybrid model's dynamic re-order points (driven by the Poisson .ppf(0.95) buffer for installed-base-linked parts and the $1.645 \times \sigma$ buffer for the rest) reduce the required inventory target for C-class parts by an estimated 40-60% relative to the current 13-weeks-of-supply assignment. That freed capital can be reallocated to buffer the high-volatility A and B parts that drive customer satisfaction. The 95% SLA is maintained or improved across the network because buffers are now sized to empirical variance rather than arbitrary heuristics.

5.6 Failure Awareness

Through simulation and expert review with Zebra's advisors, we identified two specific failure modes. First, the Poisson model produces artificially low recommendations when the upstream Installed Base data is zeroed out or unlinked in the ERP. The model behaves correctly given its inputs, but the inputs themselves are missing. Second, the Time-Series model cannot predict spikes caused by new product introductions because it relies strictly on trailing historical averages. Both failure modes are symptoms of upstream Master Data Management gaps rather than algorithmic errors, and both are addressable through the BOM cleanup work recommended in Section 7.

6. Findings and Answers to Business Questions

6.1 Question 1: Optimal Inventory Levels

By applying the 95th percentile of the Poisson demand distribution and combining it with empirical lead-time variance, the model generated mathematically optimal safety stock and re-order point values for over 5,000 critical parts across the in-scope depots, surfaced in the per-depot _COMPLETE.csv deliverables. Where installed-base linkage is missing, the time-series fallback produced a 95%-SLA-equivalent target from the part's own consumption history. Together these outputs cover the entire in-scope catalog with model-appropriate routing per part.

The benchmark depot comparison summarized in Section 5.3 (higher predictive accuracy, corrected bias, and 3× improvement in inventory turnover) quantifies the per-part benefit of moving from the current heuristic to dynamic, variance-adjusted targets.

6.2 Question 2: Maintaining CSAT at Lower Inventory

Yes. Zebra can operate at a lower network-wide inventory level while maintaining or improving the material fill rate. Two pieces of evidence support this conclusion.

First, by transitioning from static weeks-of-supply to dynamic, variance-adjusted buffers, Zebra can shrink the physical footprint currently allocated to 'C' and 'Unassigned' parts. The freed capital can be redirected to buffer high-volatility A and B parts more aggressively than the current heuristic allows, lowering total network inventory while still meeting a 95% service level. Month-over-month, our recommended targets produce lower total inventory across all products and locations while preserving the SLA.

Second, the network comparison in Section 5.4 demonstrated that higher inventory levels are correlated with more on-time shipments. The over-stocked depots were the clear best performers on customer satisfaction proxies, as the late-shipping depots were mostly on-target. Late shipping depots may likely suffer from structural issues causing late shipments, as our model predicted their current inventory levels were mostly on-target and could not be reduced further while maintaining a 95% fill rate. However,

various strong performing depots were highly overstocked, thus reductions could be made to their inventory levels without expecting a corresponding service degradation, thus it is possible to maintain current customer satisfaction levels with a lower fill rate.

6.3 Question 3: Critical Dependencies

Two operational dependencies must be addressed before Zebra can safely run at theoretical minimum inventory levels.

Internal DC-to-Depot Reliability

The binding supply-side constraint is internal, not external. Majority of depot inbound volume is internal DC-to-Depot transfers, and a portion of those internal replenishments arrive critically late, with a heavily right-skewed long tail extending past 80 days for some lanes. Strict SLA enforcement on these internal lanes is required to stabilize the network. Without it, leaner depot inventory will be exposed to the same internal variance that today is being absorbed by excess buffer stock; operating at the model's optimal levels would amplify, not mitigate, the impact of internal-transfer delays.

Master Data and BOM Linkage

Disconnected BOM data is the second binding dependency. Establishing reliable BOM linkages would unlock the Poisson model's full predictive power across the rest of the catalog.

Supplier-Side Variance

External supplier variance is a secondary but real concern. Lead-time variance from key suppliers (some exceeding 80 days) creates a ripple effect that forces depots to hold excess units to buffer the unreliability. Strict SLA enforcement with under-performing suppliers, in addition to the internal-transfer SLAs above, is required to safely lower the inventory floor at upstream Manufacturing Hubs.

Tariff Saving opportunity and COO Data Enrichment Opportunity

A notable concentration of critical spare part spend is sourced from high-tariff regions. This concentration creates material cost exposure that is difficult to quantify, manage, or mitigate without accurate (COO) data at the part level.

A meaningful portion of parts in the catalog carry COO data gap. Without this attribution, the organization lacks the visibility required to assess tariff liability by part, product family, or depot location. Decisions about sourcing alternatives, safety stock adjustments, or cost mitigation strategies cannot be made with confidence when the underlying exposure is unquantified.

Enriching this data is a prerequisite for unlocking tariff-related savings. Specifically, establishing accurate origin classification across the critical parts catalog would enable the organization to:

- Identify which parts carry elevated tariff exposure due to sourcing from high-tariff regions
- Prioritize sourcing diversification for the highest-exposure, highest-criticality parts
- Incorporate tariff-adjusted cost into replenishment target calculations
- Proactively adjust safety stock for parts at risk of supply disruption due to trade policy shifts

This analysis does not prescribe a specific sourcing outcome. Rather, it surfaces the data gap as a structural blind spot that, if addressed, would meaningfully expand the optimization model's ability to support cost-efficient and resilient inventory decisions.

6.4 Robustness and Confidence Calibration

- **Robust findings:** the inventory footprint consumed by C and Unassigned parts; the internal late-delivery rate; the 24.5 day mean internal lead time with a heavy right tail; and the Poisson-driven targets for the of parts with installed-base linkage. These are backed by historical simulation across clean records and should be acted upon.
- **Suggestive findings:** the precise inventory targets generated for parts processed through the time-series fallback. They represent a defensible historical baseline at a 95% SLA equivalent, but they should not be relied upon for proactive management of new product introductions or rapidly aging fleets until BOM linkages are established.

7. Recommendations and Next Steps

7.1 Adopt a Hybrid Phased Rollout

Rather than a wholesale replacement of the current planning system, the optimal path forward is to combine both approaches and capture the strengths of each. We recommend that the Poisson and time-series targets generated in this project be deployed first at smaller depots, where the reduction of excess inventory has the largest proportional balance-sheet impact and where leaner margins are easiest to absorb operationally. The current model can remain in place at the larger regional hubs, where its conservative targets help insulate downstream depots from upstream variance during the transition.

This phased rollout preserves Zebra's strong customer-satisfaction position while capturing inventory cost savings at the smaller locations first. It also reduces the spend associated with parts arriving too early and sitting in warehouses, the silent cost of an over-stocked, statically-buffered network.

7.2 Transition to Dynamic Re-Order Points

Supply Chain leadership should abandon static weeks-of-supply logic for 'C'-rated and 'Unassigned' items and integrate the Python-generated `optimal_inventory_level_95` logic directly into depot planning systems. Safety stock should scale dynamically with empirical variance, not arbitrary heuristics. This is the

single highest-leverage change available from the project, and it can be implemented immediately for the critical parts already covered by the model.

7.3 Target Internal Hub-to-Spoke SLAs

Operations should refocus from vendor expediting to stabilizing internal transit between global DCs and the 21 repair depots. Eliminating the internal late-delivery rate is a prerequisite to safely lowering the inventory floor; the model can size buffers correctly for any given lead-time distribution, but a more reliable distribution shifts the entire optimization toward leaner targets while preserving the same SLA.

7.4 Prioritize Master Data Management

IT and Data Strategy should prioritize mapping independent spare parts to their parent devices in the field. Establishing the missing BOM linkages will unlock the full predictive power of the Poisson model across the remaining of the catalog, replacing the conservative time-series fallback with a more responsive, installed-base-aware demand signal. We would also mathematically integrate vendor Minimum Order Quantities directly into the optimization logic in a future iteration, since rigid MOQs currently force Zebra to over-order regardless of the model's mathematically optimal lot size.

7.5 Expected Impact

If executed in this sequence, the phased rollout is expected to deliver lower month-over-month total inventory across all products and locations, while maintaining or improving the 95% material fill rate. The reduction in excess capital tied up in C and Unassigned parts will free working capital for redeployment, the bias correction will reduce the financial risk associated with systematic over-stocking, and the threefold improvement in inventory turnover observed in the benchmark depot will compound across the network as the rollout completes.

8. Limitations

8.1 Method and Data Limitations

The most consequential limitation of the hybrid approach is its dependence on the availability of linkable installed-base data. The remaining majority were routed to the time-series fallback. While the fallback is mathematically defensible at the same 95% SLA, it cannot incorporate forward-looking installed-base information and therefore cannot anticipate spikes driven by new product introductions or rapid fleet aging.

ERP data silos imposed a second class of limitation. A table looking at purchases table mixes non-inventory facility expenses (for example, office supplies) with critical spare parts, requiring programmatic filtering

to isolate the true inventory parts in scope. Similar filtering was required throughout the pipeline, and any change to the underlying ERP coding scheme would require maintenance of the cleaning rules.

8.2 Failure Modes

Two failure modes are worth re-stating because they affect how the model's outputs should be consumed. First, the Poisson model produces artificially low recommendations when upstream Installed Base data is zeroed out or unlinked in the ERP; the algorithm is correct given its inputs, but the inputs themselves are gap-driven. Second, the time-series fallback fails to anticipate spikes from new product introductions because it relies strictly on trailing averages. Both failure modes are symptoms of upstream Master Data Management gaps rather than algorithmic errors.

8.3 Implications for Confidence

Zebra leadership should place high, robust confidence in the internal lead-time standard deviations and the inventory targets for the portion of parts run through the Poisson model. Recommendations for parts forced into the time-series fallback should be treated as directionally reliable but not as drop-in replacements for active planner judgment, particularly for parts associated with newly introduced or aging product lines until BOM linkages are restored.

8.4 Future Work

With fewer constraints and additional time, the highest-impact next step is Master Data Management cleanup: specifically, building a relational mapping algorithm that reconstructs the missing BOM links between unassigned spare parts and the installed base. A second high-value extension is the explicit integration of vendor Minimum Order Quantities into the Python optimization logic, so that the recommended quantities are simultaneously mathematically optimal and operationally executable under existing vendor contracts.

9. Conclusion

We developed a hybrid inventory optimization model that produces an optimal target level for every in-scope service part at every in-scope location, using a Poisson Generalized Linear Model where installed-base linkage is available and a time-series fallback elsewhere, both sized to a 95% service-level target. Validation showed measurably lower forecast error, correction of the current model's positive overstocking bias, and a threefold improvement in inventory turnover at the benchmark depot.

Across the network, the model produces lower month-over-month total inventory while maintaining the 95% material fill rate, and the data shows clearly that higher inventory levels are not correlated with better on-time shipment performance. The overstocking embedded in Zebra's current static ABC heuristic represents a substantial pool of working capital that can be freed without compromising service.

Two operational dependencies condition the magnitude of the realized benefit. The late-delivery rate on internal DC-to-Depot transfers must be addressed first; leaner depot stock leaves less margin for absorbing internal variance, and unaddressed internal transfer delays will become the binding constraint on customer satisfaction once the buffer is removed. Second, BOM master data gaps must be closed to extend Poisson coverage across the full catalog and unlock the model's complete predictive power.

Taken together, the framework gives Zebra a defensible, data-driven path to a leaner, more responsive global service inventory: one that protects the customer-experience advantage that distinguishes the company in its market while measurably improving balance-sheet performance and reducing the silent cost of bloat. We recommend Zebra adopt the phased rollout outlined in Section 7, beginning with smaller depots and the Poisson-eligible parts, and pair it with the internal SLA enforcement and master-data cleanup that will allow the framework to be extended to the full network with confidence.

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